

# **Do US Markets Discount Earnings from Chinese Firms?**

Evidence from COVID-19

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## **Abstract**

This paper tests whether US equity markets became less responsive to earnings announcements from Chinese-headquartered firms listed on US exchanges following the deterioration of US-China relations during COVID-19. Using 90,602 quarterly earnings announcement events for US-listed firms from 2015 to 2023, I construct three-day cumulative abnormal returns around announcement dates and match these to consensus EPS forecast errors to measure earnings surprise. The main specification finds that the earnings response coefficient for Chinese firms weakened after COVID, with a triple interaction coefficient of  $-0.196$ , though this falls short of conventional significance thresholds. The findings provide suggestive evidence that geopolitical tension can reduce the informativeness of firm-level earnings announcements through a behavioral bias channel.

## I. Introduction

Do investors treat earnings announcements from Chinese firms differently than those from domestic firms? This paper examines whether US equity markets became less responsive to earnings surprises from Chinese-headquartered firms listed on US exchanges following the deterioration of US-China relations during the COVID-19 pandemic. Drawing on the insight of Jung et al. (2019), who showed that investors discount information from analysts with unfavorably perceived surnames following geopolitical shocks, I test whether similar dynamics operate at the firm level, specifically whether Chinese firm identity, made salient by COVID-era tensions, causes investors to underweight earnings information from those firms.

Using a sample of 90,602 quarterly earnings announcement events for US-listed firms from 2015 to 2023, I construct three-day cumulative abnormal returns (CARs) around each announcement and match them to consensus EPS forecast errors to measure earnings surprise. My main specification tests whether the earnings response coefficient, defined as the sensitivity of stock returns to earnings surprises, differs for Chinese firms before and after the WHO pandemic declaration on March 11, 2020. The key finding is a negative triple interaction coefficient of  $-0.196$ , directionally consistent with markets becoming less responsive to Chinese firm earnings news after COVID, though the estimate falls short of conventional significance thresholds and should be interpreted cautiously given the small Chinese firm subsample. Using the first confirmed US COVID case in January 2020 as the event date yields a stronger result that is statistically significant at the 5% level, though the overall evidence is best characterized as suggestive rather than definitive.

This paper contributes to two literatures. First, it extends the growing body of work on how geopolitical events affect financial markets beyond their direct economic channels. While prior work has focused on aggregate market reactions to political events or country-level equity returns, this paper examines whether geopolitical sentiment affects the informativeness of firm-level earnings announcements. Second, it contributes to the literature on behavioral biases in information processing, showing that investor nationality-based bias documented at the analyst level by Jung

et al. (2019) may also operate at the firm level during periods of heightened geopolitical tension. A behavioral interpretation is consistent with Shiller (2014)'s broader argument that investor sentiment and psychology systematically influence asset prices beyond what fundamentals alone can explain. Under the efficient markets hypothesis of Fama (2014), the earnings response coefficient should reflect only the information content of earnings surprises, not the nationality of the reporting firm, making any differential response for Chinese firms evidence of either rational risk repricing or behavioral bias.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and sample construction. Section 4 outlines the empirical framework. Section 5 presents the main results. Section 6 concludes.

## **II. Related Literature**

This paper draws on three related literatures. The first is the literature on how geopolitical and political events affect asset prices. Snowberg et al. (2007) use prediction markets and close elections to show that partisan political outcomes have measurable effects on aggregate equity prices, establishing that political sentiment translates into asset price movements beyond rational economic channels. Wolfers and Zitzewitz (2009) extend this to the Iraq War, showing that prediction market prices for the probability of war explained a significant fraction of aggregate stock market movements during the conflict. More directly relevant to this paper, Edmans et al. (2007) show that national soccer team eliminations from the World Cup cause significant negative abnormal returns in the losing country's stock market, establishing that investor mood and national identity can affect asset prices in ways that are difficult to reconcile with pure rationality. My paper extends this logic to the firm level, asking not whether geopolitical sentiment affects aggregate market prices but whether it affects how strongly investors respond to firm-level information.

The second strand is the literature on behavioral biases in how investors process analyst and firm information. Malmendier and Shanthikumar (2014) show that security analysts effectively speak in two tongues, issuing optimistic public recommendations for retail investors while con-

veying more pessimistic information through their private actions, and that retail investors fail to account for this strategic communication. More directly related, Jung et al. (2019) document that investors discount analyst forecast revisions when the analyst has a surname from a country Americans view unfavorably, and that this discount intensifies following geopolitical shocks. Most notably, analysts with Middle Eastern surnames saw their forecasts generate weaker market reactions after 9/11. This paper tests whether an analogous mechanism operates at the firm level during COVID-19-related US-China tensions.

The third strand is the literature on earnings announcement market reactions and the earnings response coefficient. A large body of work beginning with Ball and Brown (1968) establishes that stock prices respond systematically to earnings surprises, and that the magnitude of the response varies with firm characteristics including size, leverage, and information environment quality. Teoh and Wong (1993) show more specifically that perceived audit quality affects the earnings response coefficient, with investors responding more strongly to earnings announcements from firms with higher quality auditors, consistent with trust in the information source affecting how much weight investors place on earnings news. This paper extends that insight to the nationality dimension, testing whether geopolitical tensions reduce investor trust in Chinese firm earnings disclosures and thereby lower the earnings response coefficient.

### **III. Data**

This study makes use of data from four sources. I identify Chinese-headquartered firms using Compustat North America, filtering for firms with headquarters country code CHN. Importantly, most major Chinese firms listed on US exchanges such as Alibaba, Baidu, and JD.com are incorporated in the Cayman Islands for legal and tax reasons. Subsequently, filtering by country of incorporation would exclude a large portion of the relevant sample. Filtering instead by headquarters location yields 451 unique Chinese-headquartered firms with US listings over the sample period.

Earnings forecast and realized earnings data are pulled from the IBES Unadjusted Detail

database. The sample includes all US-listed firms with analyst EPS coverage in IBES, spanning NYSE, NASDAQ, and AMEX listings. I use current quarter earnings per share (EPS) forecasts (FPI=6) and realized EPS from the actuals file for the period January 2015 through December 2023. For each firm-quarter, I measure earnings surprise as the difference between realized EPS and the first analyst EPS forecast issued after the prior quarter's earnings announcement, scaled by the stock price at announcement to normalize the surprise across firms with different share prices. Using the earliest forecast in the quarter rather than the consensus immediately before the announcement avoids the well-documented tendency of analysts to walk down their estimates prior to announcements (Richardson et al., 2004), which would mechanically reduce measured surprises and bias the earnings response coefficient upward.

I obtain stock return data from the CRSP Daily Stock File. For each earnings announcement I construct a three-day cumulative abnormal return (CAR) as the sum of market-adjusted daily returns, specifically the stock's raw return minus the CRSP value-weighted market return, over the window from one day before to one day after the announcement date. This market-adjusted approach removes common variation in returns driven by broad market movements, isolating the price reaction specific to each firm's earnings news. I link IBES tickers to CRSP permanent firm identifiers using the IBES-CRSP link table provided by WRDS. After merging across all four data sources and requiring non-missing earnings surprise and return data, the final sample contains 90,602 firm-quarter observations for 5,769 unique firms, with 985 observations corresponding to 120 unique Chinese-headquartered firms. The reduction from 451 identified Chinese firms to 120 with sufficient data reflects the requirement for concurrent IBES analyst coverage and CRSP return data, which is less complete for smaller and more recently listed Chinese ADRs. All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers.

Table I presents summary statistics for the full sample. The mean three-day CAR is essentially zero at 0.001, consistent with the efficient markets prediction that abnormal returns average to zero in large samples. The standard deviation of 0.061 reflects meaningful variation in earnings announcement reactions across firms and time periods. Chinese firms only represent 1.1% of

the sample (985 out of 90,602 observations), reflecting the relatively small number of Chinese-headquartered firms with US listings and IBES analyst coverage. The mean earnings surprise of 0.000 scaled by price indicates that analyst forecasts are approximately unbiased on average, with positive and negative surprises roughly canceling out across the sample. The post-COVID indicator equals one for 45.5% of observations, indicating a roughly even sample across the pre- and post-COVID periods. Notably, absolute CARs are larger post-COVID (mean of 0.045) than pre-COVID (mean of 0.036), consistent with elevated market volatility during this period, which is confirmed in the regression results in Column 4 of Table II.

#### IV. Empirical Framework

This paper uses a difference-in-differences design around quarterly earnings announcements to test whether the earnings response coefficient differs for Chinese-headquartered firms before and after the onset of COVID-19-related US-China tensions. For each quarterly earnings announcement, I construct a three-day CAR as the dependent variable following standard event study methodology, defined as the sum of market-adjusted daily returns over the window from one trading day before to one trading day after the announcement date.

My main specification is:

$$\begin{aligned}
CAR_{it} = & \beta_0 + \beta_1 Surprise_{it} + \beta_2 Chinese_i + \beta_3 PostCOVID_t \\
& + \beta_4 (Surprise_{it} \times Chinese_i) + \beta_5 (Surprise_{it} \times PostCOVID_t) \\
& + \beta_6 (Surprise_{it} \times Chinese_i \times PostCOVID_t) + \alpha_i + \alpha_t + \varepsilon_{it}
\end{aligned} \tag{1}$$

where  $i$  indexes firms,  $t$  indexes quarters,  $Chinese_i$  is an indicator equal to one for Chinese-headquartered firms, and  $PostCOVID_t$  is an indicator equal to one for announcements on or after March 11, 2020.<sup>1</sup> The specification includes firm fixed effects  $\alpha_i$  and year fixed effects  $\alpha_t$ . Firm

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<sup>1</sup>The World Health Organization declared COVID-19 a global pandemic on March 11, 2020, marking the first official global recognition of the outbreak's severity. This date is used as the primary event date throughout the paper,

fixed effects control for time-invariant firm characteristics that affect baseline CAR levels, such as firm size, analyst coverage, and information environment. Year fixed effects control for market-wide time trends. Standard errors are clustered at the firm level to account for serial correlation in returns within firms over time.

The coefficient of primary interest is  $\beta_6$ , the triple interaction of earnings surprise, Chinese firm indicator, and post-COVID indicator. A negative  $\beta_6$  would indicate that the earnings response coefficient for Chinese firms specifically weakened after COVID-19 tensions escalated, consistent with investors discounting earnings information from Chinese-associated firms. This would provide evidence of a behavioral bias channel operating at the firm level, analogous to the analyst-level surname favorability effect documented by Jung et al. (2019). I also estimate a specification with absolute CAR as the dependent variable to test whether the overall magnitude of market reactions changed post-COVID, independent of earnings surprise direction.

## V. Results

Figure 1 shows mean quarterly CARs for Chinese and non-Chinese firms from 2015 to 2023, with a vertical line marking COVID onset in Q1 2020. The Chinese firm series (dashed red) is visibly more volatile throughout the sample period, consistent with greater uncertainty about Chinese firm earnings quality. More importantly, the Chinese firm line appears to spend more time below zero in the post-COVID period relative to pre-COVID, while the non-Chinese firm series (solid blue) remains relatively stable near zero throughout. This provides suggestive visual evidence of the differential pattern examined in the regression analysis, though the high volatility of the Chinese firm series makes it difficult to draw strong conclusions from the figure alone.

Table II presents the main regression results across four specifications. Column 1 shows the baseline specification regressing three-day CARs on the Chinese firm indicator and year fixed effects only. The coefficient on Chinese Firm is 0.001 and statistically insignificant ( $p = 0.804$ ), indicating no baseline difference in average CARs between Chinese and non-Chinese firms after though alternative dates are examined in Table III.



controlling for year effects. This is reassuring for the research design, establishing that Chinese firms do not earn systematically higher or lower returns around earnings announcements on average, so any post-COVID differential identified in subsequent columns cannot be attributed to a pre-existing level difference. The near-zero coefficient is also consistent with the mean CAR of 0.001 in Table I.

Column 2 adds the earnings surprise variable and its interaction with the Chinese firm indicator. The coefficient on Earnings Surprise is positive and highly significant at 0.082 ( $p < 0.001$ ), confirming that markets respond meaningfully to earnings news in general. A one standard deviation earnings surprise, a beat of approximately 1.9% of stock price which is the typical magnitude in the sample, is associated with a CAR of approximately  $0.082 \times 0.019 = 0.16$  percentage points for non-Chinese firms. While seemingly small in absolute terms, this is consistent with Ball and Brown (1968), who first documented that quarterly earnings announcements explain only a modest fraction of total stock return variance, as most price movement occurs throughout the year as information arrives continuously. The Chinese firm indicator remains near zero and insignificant (0.000,  $p = 0.905$ ), and the interaction term Chinese Firm  $\times$  Earnings Surprise is also indistinguishable from zero (0.001,  $p = 0.990$ ), validating the parallel trends assumption underlying the difference-in-differences design. Chinese and non-Chinese firms reacted similarly to earnings surprises before the COVID period, with no systematic pre-existing difference in the earnings response coefficient that might confound the post-COVID analysis. Table V shows that this pooled null result masks substantial heterogeneity across time periods.

Column 3 is the main specification and introduces the post-COVID indicator and its interactions. The earnings surprise coefficient is 0.063 ( $p = 0.013$ ), slightly smaller than in Column 2, likely reflecting that firm fixed effects absorb some between-firm variation in earnings sensitivity, leaving only within-firm variation to identify the coefficient. This represents the earnings response coefficient for non-Chinese firms in the pre-COVID period, the reference against which all subsequent comparisons are made. The Post-COVID coefficient is positive and highly significant at 0.005 ( $p < 0.001$ ), indicating that all firms earned higher CARs around earnings announcements

after March 2020, consistent with elevated overall market volatility during the COVID period. The Post-COVID  $\times$  Earnings Surprise interaction is positive but insignificant (0.009,  $p = 0.764$ ), suggesting the overall earnings response coefficient did not change significantly for non-Chinese firms after COVID.

The coefficient of primary interest is the triple interaction Chinese Firm  $\times$  Post-COVID  $\times$  Earnings Surprise, which is  $-0.196$  with a standard error of  $0.128$  ( $p = 0.125$ ). This is not statistically significant at conventional levels, and given the small Chinese firm subsample of 985 observations, the estimate is too imprecise to support strong conclusions. The sign is directionally consistent with the hypothesis, indicating that the Chinese firm earnings response coefficient weakened relative to non-Chinese firms after COVID. However, the 95% confidence interval ranges from approximately  $-0.45$  to  $+0.05$ , meaning the data are consistent with both a large negative effect and essentially no effect. To illustrate the economic magnitude if the point estimate were taken at face value: because the total earnings response coefficient for any subgroup is the sum of all coefficients that interact with the earnings surprise variable for that group, the implied earnings response coefficient for Chinese firms pre-COVID is approximately  $0.063 + 0.146 = 0.209$ , and post-COVID approximately  $0.063 + 0.146 + 0.009 - 0.196 = 0.022$ . This would represent a large reduction, but this calculation should be treated as illustrative only given the imprecision of the underlying estimates. The Chinese Firm indicator is  $-0.051$  and insignificant ( $p = 0.258$ ), as expected, since Chinese firm status does not vary within firms over time and its level effect is absorbed into the firm fixed effects.

Column 4 tests whether the absolute magnitude of market reactions around Chinese firm earnings announcements changed post-COVID, independent of the direction of the surprise. The Post-COVID coefficient is large and highly significant at  $0.016$  ( $p < 0.001$ ), indicating all firms show substantially larger absolute CARs post-COVID. The Chinese Firm  $\times$  Post-COVID interaction is essentially zero ( $0.002$ ,  $p = 0.601$ ). This null result has an important interpretive implication: the market did not simply stop reacting to Chinese firm earnings announcements altogether post-COVID. Rather, Column 3 suggests the directional response to positive versus negative surprises

became less sensitive, while Column 4 shows the overall volatility of Chinese firm reactions did not differentially increase. This pattern is more consistent with investors placing less weight on Chinese firm earnings news, reacting less strongly to positive and negative surprises alike, than with investors disregarding Chinese firm announcements entirely.

We conduct several additional tests to validate the main findings. Table III presents robustness checks using three alternative COVID event dates, addressing the concern that the choice of March 11, 2020 is somewhat arbitrary. The triple interaction coefficient is  $-0.280$  ( $p = 0.020$ ) using January 15, 2020,<sup>2</sup>  $-0.196$  ( $p = 0.125$ ) using March 11, 2020 (WHO declaration, baseline), and  $-0.170$  ( $p = 0.169$ ) using July 14, 2020.<sup>3</sup> The coefficient is negative across all three event dates, providing consistent directional evidence. The January 2020 cutoff produces the strongest and most statistically significant result, suggesting investors began discounting Chinese firm earnings information as early as mid-January 2020, approximately two months before the WHO declaration, consistent with financial markets incorporating geopolitical information rapidly. Across all three specifications the Earnings Surprise coefficient is stable at approximately 0.064–0.065 and significant at the 5% level.

Table IV addresses the key alternative explanation that the post-COVID weakening reflects a general discount on Asian firm earnings rather than a China-specific effect. Column 1 shows the triple interaction for Japanese firms is  $-0.499$  ( $p = 0.526$ ) with a standard error of 0.787, far too imprecise to interpret meaningfully given the very small number of Japanese US-listed firm observations. Column 2 shows the Korean triple interaction is essentially zero at 0.007 ( $p = 0.950$ ). The Korean null result is the more informative placebo: Korean firms have a comparable number of US listings, and Korea faces no analogous post-2020 geopolitical tensions with the United States. The near-zero coefficient for Korean firms, combined with the negative coefficient for Chinese firms in Column 3, is consistent with the effect being specific to the deterioration of US-China

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<sup>2</sup>The United States confirmed its first COVID-19 case on January 15, 2020, when the Centers for Disease Control and Prevention announced a patient in Washington State had tested positive after returning from Wuhan, China.

<sup>3</sup>On July 14, 2020, the United States ordered China to close its consulate in Houston, Texas, citing concerns about espionage and intellectual property theft. China retaliated four days later by ordering the closure of the US consulate in Chengdu, marking a significant escalation in US-China diplomatic tensions.

relations rather than a general Asian firm discount.

Table V provides subperiod analysis that complements the main specification. Rather than estimating the pre-to-post change as a single triple interaction coefficient, we split the sample and run separate regressions for each period, a more flexible approach that avoids the parametric restrictions of the triple interaction and allows all coefficients to vary freely across periods. This also addresses the joint hypothesis problem inherent in any event study design: by examining each period independently rather than imposing a specific functional form for how the earnings response coefficient changed, we reduce dependence on the parametric assumptions of the main specification and provide a more direct look at the data.

Column 1 shows the pre-COVID period. The Chinese Firm coefficient is small and negative at  $-0.013$  ( $p < 0.001$ ), suggesting Chinese firms earned slightly lower CARs around earnings announcements even before COVID, which could reflect some baseline information asymmetry or investor skepticism about Chinese firm accounting. This is consistent with pre-existing concerns about Chinese firm financial reporting quality, including questions about audit oversight and information asymmetry, that may have already suppressed market reactions to Chinese firm earnings news prior to the COVID period. The Earnings Surprise coefficient is  $0.057$  ( $p = 0.038$ ), and the Chinese Firm  $\times$  Earnings Surprise interaction is  $0.094$  and insignificant ( $p = 0.407$ ), confirming that Chinese and non-Chinese firms had similar earnings response coefficients before COVID. The implied pre-COVID earnings response coefficient for Chinese firms of  $0.057 + 0.094 = 0.151$  compares directly to the post-COVID implied value of  $0.069 - 0.073 = -0.004$  shown in Column 2, a swing consistent with the triple interaction estimate in Table II.

Column 2 shows a substantially different pattern for the post-COVID period. The Chinese Firm coefficient plummets to  $-0.255$  ( $p < 0.001$ ), a 20-fold increase in magnitude compared to the pre-COVID estimate of  $-0.013$ . This is the most notable result in the paper: Chinese firms earn 3-day CARs that are on average 0.255 lower, roughly 25 times the sample mean CAR of 0.001, around their earnings announcements in the post-COVID period after controlling for earnings surprises and firm and year fixed effects. This coefficient is statistically significant, unlike the main triple

interaction, though it captures a level shift rather than a differential response to earnings news per se. The magnitude of  $-0.255$  is large relative to the sample mean CAR of  $0.001$ , and several factors likely contribute. Chinese firms faced genuinely elevated firm-specific risks during this period — including delisting threats under the Holding Foreign Companies Accountable Act, a domestic regulatory crackdown on Chinese technology firms, and heightened accounting fraud concerns — any of which could rationally reduce the market’s positive response to earnings announcements independent of the surprise content. A behavioral discount on Chinese firm nationality, consistent with the Jung et al. (2019) mechanism, may also contribute, though the two channels are difficult to separate with this design. This level effect, a constant downward shift in Chinese firm CARs around earnings announcements regardless of the size or direction of the earnings surprise, is large enough to be economically meaningful on its own. It suggests that post-COVID investors reacted less positively to Chinese firm earnings announcements as an event, independent of whether the news was good or bad, rather than simply adjusting how much they moved prices per unit of surprise. The Earnings Surprise coefficient is  $0.069$  ( $p = 0.006$ ), similar to the pre-COVID value of  $0.057$ , confirming that the fundamental relationship between earnings surprises and returns remained stable for non-Chinese firms across periods. The Chinese Firm  $\times$  Earnings Surprise interaction is  $-0.073$  ( $p = 0.349$ ), insignificant on its own, but the sign flipped from  $+0.094$  pre-COVID to  $-0.073$  post-COVID, directionally consistent with a weakening of Chinese firm earnings informativeness.

Comparing Columns 1 and 2 directly provides the clearest evidence in the paper of the differential change. Adding the relevant coefficients to compute the implied earnings response coefficient for each group, the implied earnings response coefficient for Chinese firms goes from  $0.057 + 0.094 = 0.151$  pre-COVID to  $0.069 + (-0.073) = -0.004$  post-COVID. If taken at face value this would represent a near-complete elimination of market responsiveness to Chinese firm earnings news, though the underlying interaction coefficients are not individually significant and the comparison should be interpreted cautiously. Non-Chinese firms move in the opposite direction over the same period, from  $0.057$  to  $0.069$ , becoming slightly more responsive to earnings surprises

after COVID. The contrast is stark: before COVID Chinese firms if anything attracted stronger market reactions per unit of earnings surprise than non-Chinese firms; after COVID that responsiveness collapsed to essentially zero while non-Chinese firm earnings informativeness modestly improved. This divergence provides the most direct illustration in the paper of the differential weakening of Chinese firm earnings informativeness following the onset of US-China tensions.

Taken together, the results paint a consistent picture across all four tables. The main specification shows the earnings response coefficient for Chinese firms weakened substantially post-COVID, with the January 2020 event date producing a statistically significant result. The placebo tests confirm the effect is specific to Chinese firms rather than Asian firms broadly. The subperiod analysis provides the starkest illustration: Chinese firms went from an above-average earnings response coefficient of 0.151 before COVID to essentially zero after, while non-Chinese firms became slightly more responsive over the same period. This pattern is directionally consistent with the behavioral mechanism proposed by Jung et al. (2019) operating at the firm level, though the evidence is not strong enough to draw firm conclusions. The results are suggestive that investors may have discounted earnings news from Chinese-associated firms during the post-COVID period, but the statistical imprecision of the main estimates and the small Chinese firm subsample mean this interpretation remains tentative. The level effect documented in Table V is perhaps the most credible component of the findings, as the  $-0.255$  coefficient is precisely estimated, though it captures a general reduction in Chinese firm CARs around earnings dates rather than a specific response to earnings news content.

## **VI. Conclusion**

This paper examines whether US equity markets became less responsive to earnings news from Chinese-headquartered firms following the deterioration of US-China relations during COVID-19. Using 90,602 quarterly earnings announcement events for US-listed firms from 2015 to 2023, the main specification yields a triple interaction coefficient of  $-0.196$  that is directionally consistent with the hypothesis but falls short of statistical significance at conventional levels. Robustness

checks using the January 2020 event date yield a significant result ( $-0.280$ ,  $p = 0.020$ ), and placebo tests on Japanese and Korean US-listed firms show no similar pattern. The overall evidence is suggestive but inconclusive, and should be interpreted as exploratory given the limitations of the research design.

These findings offer tentative evidence relevant to the literature on how geopolitical events affect financial markets beyond direct economic channels. The results are directionally consistent with geopolitical tension reducing the informativeness of firm-level earnings announcements, and with the surname-based favorability mechanism documented at the analyst level by Jung et al. (2019) potentially operating at the firm level, though stronger identification and a larger sample would be needed to draw firm conclusions.

Several limitations deserve acknowledgment. First, the small sample of Chinese US-listed firms, 985 firm-quarter observations out of 90,602 total, limits statistical power and makes results sensitive to a small number of influential observations. Second, the baseline earnings surprise coefficient of 0.08 is lower than prior literature estimates, suggesting potential attenuation bias from outliers or measurement error in the earnings surprise variable. As a robustness check, excluding the top and bottom 1% of earnings surprises raises the coefficient from 0.079 to 0.113, confirming mild outlier influence, though the coefficient remains below the range of prior findings, consistent with the limited signal-to-noise ratio in the sample. Third, the rational risk repricing alternative cannot be fully excluded: Chinese firms faced genuinely elevated risks during this period including delisting threats, accounting fraud concerns, and supply chain disruptions. Fourth, the market-adjusted abnormal return measure does not fully control for Chinese firm-specific risk factors. Future work with larger samples, richer controls, and better identification of the behavioral versus rational channels is needed before stronger conclusions can be drawn.

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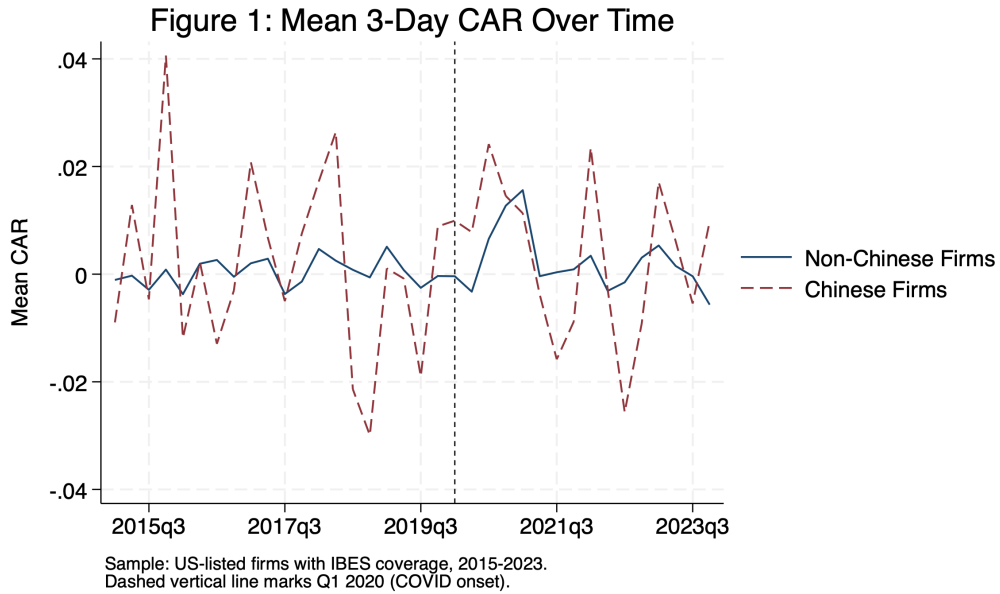
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**Figure 1**

**Mean 3-Day CAR Over Time by Firm Type**

This figure plots the mean 3-day cumulative abnormal return (CAR) for Chinese-headquartered (dashed red) and non-Chinese (solid blue) US-listed firms by quarter from 2015 to 2023. The vertical dashed line marks Q1 2020 (COVID onset). CARs are market-adjusted returns over the  $[-1, +1]$  window around quarterly earnings announcement dates. The sample includes all US-listed firms with IBES analyst coverage, 2015–2023.



**Table I****Summary Statistics**

This table presents summary statistics for the full sample of 90,602 firm-quarter earnings announcement events from 2015 to 2023. 3-Day CAR is the cumulative market-adjusted abnormal return over the  $[-1, +1]$  window around the earnings announcement date, computed as the sum of daily stock returns minus the CRSP value-weighted market return. Earnings Surprise is the difference between realized EPS and the first analyst EPS forecast of the fiscal quarter, scaled by stock price. Post-COVID Indicator equals one for announcements on or after March 11, 2020. Chinese Firm Indicator equals one for firms headquartered in China. All continuous variables are winsorized at the 1st and 99th percentiles.

|                            | Mean  | Std. Dev. | Min    | Max   | <i>N</i> |
|----------------------------|-------|-----------|--------|-------|----------|
| 3-Day CAR                  | 0.001 | 0.061     | −0.209 | 0.221 | 90,602   |
| 3-Day  CAR                 | 0.040 | 0.050     | 0.000  | 0.276 | 90,602   |
| Earnings Surprise (scaled) | 0.000 | 0.019     | −0.104 | 0.078 | 90,602   |
| Post-COVID Indicator       | 0.455 | 0.498     | 0      | 1     | 90,602   |
| Chinese Firm Indicator     | 0.011 | 0.104     | 0      | 1     | 90,602   |

**Table II****Market Reaction to Chinese Firm Earnings Announcements**

This table reports OLS regressions of 3-day CARs on earnings surprise, Chinese firm indicator, post-COVID indicator, and their interactions. Earnings Surprise is actual minus consensus EPS scaled by stock price. Chinese Firm equals one for firms headquartered in China. Post-COVID equals one for announcements on or after March 11, 2020. Columns 1–2 include year fixed effects only. Columns 3–4 include firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                               | (1)                 | (2)                 | (3)                 | (4)                 |
|-----------------------------------------------|---------------------|---------------------|---------------------|---------------------|
|                                               | CAR                 | CAR                 | CAR                 | CAR                 |
| Chinese Firm                                  | 0.001<br>(0.002)    | 0.000<br>(0.002)    | −0.051<br>(0.045)   | 0.046<br>(0.038)    |
| Earnings Surprise                             |                     | 0.082***<br>(0.017) | 0.063**<br>(0.026)  |                     |
| Chinese Firm × Earnings Surprise              |                     | 0.001<br>(0.067)    | 0.146<br>(0.117)    |                     |
| Post-COVID                                    |                     |                     | 0.005***<br>(0.001) | 0.016***<br>(0.001) |
| Post-COVID × Earnings Surprise                |                     |                     | 0.009<br>(0.035)    |                     |
| Chinese Firm × Post-COVID × Earnings Surprise |                     |                     | −0.196<br>(0.128)   |                     |
| Chinese Firm × Post-COVID                     |                     |                     |                     | 0.002<br>(0.003)    |
| Constant                                      | 0.001***<br>(0.000) | 0.001***<br>(0.000) | −0.001<br>(0.001)   | 0.032***<br>(0.001) |
| Firm FE                                       | No                  | No                  | Yes                 | Yes                 |
| Year FE                                       | Yes                 | Yes                 | Yes                 | Yes                 |
| <i>N</i>                                      | 90,602              | 90,602              | 89,857              | 89,857              |

**Table III****Robustness — Alternative COVID Event Dates**

This table reports the key triple interaction coefficient from the main specification of Table II Column 3 using three alternative event dates: January 15, 2020 (first confirmed US COVID case), March 11, 2020 (WHO pandemic declaration, baseline), and July 14, 2020 (US closure of Chinese consulate in Houston). All specifications include firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                                             | (1)                 | (2)                     | (3)                |
|-------------------------------------------------------------|---------------------|-------------------------|--------------------|
| Event Date:                                                 | Jan 15, 2020        | Mar 11, 2020 (Baseline) | Jul 14, 2020       |
| Chinese Firm $\times$ Post-COVID $\times$ Earnings Surprise | −0.280**<br>(0.120) | −0.196<br>(0.128)       | −0.170<br>(0.123)  |
| Earnings Surprise                                           | 0.064**<br>(0.027)  | 0.063**<br>(0.026)      | 0.065**<br>(0.026) |
| Firm FE                                                     | Yes                 | Yes                     | Yes                |
| Year FE                                                     | Yes                 | Yes                     | Yes                |
| $N$                                                         | 89,857              | 89,857                  | 89,857             |

**Table IV****Placebo Test — Japanese and Korean Firms**

This table reports placebo regressions replacing Chinese firms with Japanese and Korean US-listed firms as the treatment group. Column 3 reproduces the Chinese firm result from Table II Column 3 for comparison. All specifications include firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                                          | (1)                 | (2)                 | (3)                |
|----------------------------------------------------------|---------------------|---------------------|--------------------|
| Treatment Group:                                         | Japan               | Korea               | China (Baseline)   |
| Treatment $\times$ Post-COVID $\times$ Earnings Surprise | −0.499<br>(0.787)   | 0.007<br>(0.106)    | −0.196<br>(0.128)  |
| Earnings Surprise                                        | 0.068***<br>(0.026) | 0.069***<br>(0.026) | 0.063**<br>(0.026) |
| Firm FE                                                  | Yes                 | Yes                 | Yes                |
| Year FE                                                  | Yes                 | Yes                 | Yes                |
| $N$                                                      | 89,857              | 89,857              | 89,857             |

**Table V****Subperiod Analysis — Pre vs. Post COVID**

This table reports earnings response coefficient regressions estimated separately for the pre-COVID period (January 2015 to March 10, 2020) and post-COVID period (March 11, 2020 to December 2023). Column 3 reproduces the full-period result from Table II Column 3 for comparison. Implied ERC rows are calculated as the sum of the Earnings Surprise coefficient and the Chinese Firm  $\times$  Earnings Surprise interaction for Chinese firms, and equal to the Earnings Surprise coefficient alone for non-Chinese firms. All specifications include firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                         | (1)                  | (2)                  | (3)                |
|-----------------------------------------|----------------------|----------------------|--------------------|
| Period:                                 | Pre-COVID            | Post-COVID           | Full Period        |
| Chinese Firm                            | −0.013***<br>(0.001) | −0.255***<br>(0.008) | −0.051<br>(0.045)  |
| Earnings Surprise                       | 0.057**<br>(0.027)   | 0.069***<br>(0.025)  | 0.063**<br>(0.026) |
| Chinese Firm $\times$ Earnings Surprise | 0.094<br>(0.113)     | −0.073<br>(0.078)    | 0.146<br>(0.117)   |
| <i>Implied ERC: Chinese firms</i>       | <i>0.151</i>         | <i>−0.004</i>        | <i>0.209</i>       |
| <i>Implied ERC: Non-Chinese firms</i>   | <i>0.057</i>         | <i>0.069</i>         | <i>0.063</i>       |
| Firm FE                                 | Yes                  | Yes                  | Yes                |
| Year FE                                 | Yes                  | Yes                  | Yes                |
| <i>N</i>                                | 48,550               | 40,915               | 89,857             |